

# 12.579

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## Question:

Let  $V$  be the vector space of all  $3 \times 3$  matrices with complex entries over the real field. If

$$W_1 = \{\mathbf{A} \in V : \mathbf{A} = \bar{\mathbf{A}}^\top\}$$

$$W_2 = \{\mathbf{A} \in V : \text{trace}(\mathbf{A}) = 0\}$$

then the dimension of  $W_1 + W_2$  is equal to \_\_\_\_\_.

## Solution:

The dimension of the sum of two subspaces is given by the formula:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) \quad (1)$$

The dimension of  $V$ , the space of  $3 \times 3$  complex matrices over  $\mathbb{R}$ , is:

$$\dim(V) = 3 \times 3 \times 2 = 18 \quad (2)$$

For  $W_1$  (Hermitian matrices), the dimension is determined by 3 real diagonal elements and 3 complex upper-triangular elements.

$$\dim(W_1) = 3(1) + 3(2) = 9 \quad (3)$$

For  $W_2$  (trace-zero matrices), the condition imposes two real constraints on the dimension of  $V$ .

$$\dim(W_2) = \dim(V) - 2 = 18 - 2 = 16 \quad (4)$$

For  $W_1 \cap W_2$  (Hermitian, trace-zero matrices), the trace condition imposes one real constraint on the dimension of  $W_1$ .

$$\dim(W_1 \cap W_2) = \dim(W_1) - 1 = 9 - 1 = 8 \quad (5)$$

Substituting these values into the dimension sum formula:

$$\dim(W_1 + W_2) = 9 + 16 - 8 = 17 \quad (6)$$

The dimension of  $W_1 + W_2$  is 17.